

Cobalt Strike Beacon

Project Requirements & Documentation

Scope: Post-exploitation activity, command-and-control communication, lateral movement simulation, and threat detection analysis

1. Executive Summary

This document provides an overview of Cobalt Strike Beacon as an advanced post-exploitation and adversary simulation tool used in modern cybersecurity operations. The purpose of Beacon is to help security professionals simulate real-world cyberattacks, analyze attacker behaviour, and evaluate organizational defense mechanisms in controlled environments. Humans built a legal hacking framework so defenders can experience stress in a safe educational format. Like fire drills, except the fire is ransomware and everyone suddenly remembers they ignored patch management for six months.

Beacon functions as the primary payload within Cobalt Strike and enables communication between compromised systems and command-and-control (C2) infrastructure. It supports multiple communication protocols and allows operators to perform activities such as privilege escalation, persistence, credential dumping, lateral movement, and data exfiltration. The platform is widely used during penetration testing, red team assessments, and threat emulation exercises to improve detection and incident response capabilities.

The topic also covers how threat actors misuse leaked or cracked versions of Cobalt Strike Beacon in ransomware campaigns, phishing operations, and advanced persistent threat (APT) attacks. Due to its dual-use nature, Beacon has become one of the most monitored and detected offensive security tools in enterprise environments.

2. Scope of the Topic

- Understanding Cobalt Strike architecture and Beacon functionality
- Post-exploitation techniques and attacker simulation
- Command-and-control (C2) communication methods
- Privilege escalation and lateral movement techniques
- Beacon persistence and evasion mechanisms
- Threat detection and incident response strategies

- MITRE ATT&CK mapping of Beacon activities
- Analysis of Beacon misuse in ransomware and cybercrime operations
- Network traffic monitoring and IOC identification
- Defensive security measures against Beacon-based attacks

3. Core Functional Modules

Module	Purpose	Key Outputs
Beacon Payload Engine	Maintains communication with target systems	Active sessions, task execution
Command & Control Module	Handles encrypted C2 communication	HTTP, HTTPS, DNS, SMB traffic
Post-Exploitation Toolkit	Executes attack simulation activities	Credential access, persistence
Lateral Movement Module	Simulates movement across systems	Remote execution activity
Threat Detection Layer	Identifies suspicious Beacon behaviour	Alerts, IOC logs
Reporting & Analysis Module	Documents attack simulation results	Security assessment reports

4. Beacon Intelligence Data Model

Each Beacon activity record should support the following fields:

- Beacon session ID and host information
- IP address and domain communication records
- Communication protocol used (HTTP, HTTPS, DNS, SMB)
- Timestamp of first-seen and last-seen activity
- Privilege escalation attempts
- Persistence techniques observed
- Executed commands and task history

- Associated malware or ransomware campaigns
- MITRE ATT&CK mapped TTPs
- Indicators of compromise (IOCs)
- Detection signatures and YARA references
- Network traffic characteristics and anomalies
- Related threat actor activity where applicable

5. Attack Simulation Tracking Fields

Session ID	Target System	Activity Period	Technique Used	IOC Summary	Detection Notes
BCN-001	Windows Host	First Seen ~ Last Seen	T1059, T1021	Domains, hashes, IPs	Beacon traffic identified

6. Beacon Behavioural Documentation

Every Beacon analysis page should capture:

- Beacon communication behavior and traffic patterns
- Supported command-and-control channels
- Encryption and obfuscation methods
- Process injection and memory execution behavior
- Credential dumping techniques
- Persistence and privilege escalation activity
- Common attacker commands and scripts
- Associated ransomware or APT usage
- Network indicators and suspicious connections
- Detection rules, signatures, and YARA logic
- Endpoint artifacts and forensic traces
- Links to threat reports and security references

7. Threat Detection & Defensive Monitoring

Beacon detection workflows should focus on identifying abnormal communication patterns, unauthorized persistence mechanisms, suspicious PowerShell execution, unusual SMB traffic, and encrypted outbound connections to unknown infrastructure. Detection systems should include endpoint monitoring, SIEM correlation, IDS/IPS analysis, and behavioural analytics.

- Beacon traffic analysis and anomaly detection
- Detection of malicious PowerShell and scripting activity
- Monitoring suspicious DNS tunneling behaviour
- Identification of lateral movement attempts
- Memory analysis and process injection detection
- IOC correlation with known threat intelligence feeds
- Incident response and containment procedures
- Threat hunting using MITRE ATT&CK techniques

8. Final Deliverables

- Beacon analysis documentation
- Command-and-control traffic reports
- IOC and threat intelligence database
- MITRE ATT&CK mapping reports
- Detection rules and YARA signatures
- Incident response recommendations
- Threat hunting and monitoring reports
- Post-exploitation activity analysis documentation